

Progress-Aware Multimodal Autonomy for Robotic Bronchoscopy: Recurrent Imitation Learning, Visual Junction Guidance, and Continuous Suction

Anonymous Authors

Abstract—Autonomous bronchoscopy requires long-horizon motor coordination, reliable anatomical localization, local centering at visually ambiguous bifurcations, and uninterrupted secretion management. We present a three-layer multimodal autonomy framework learned from 87 real-robot expert demonstrations containing 30,730 synchronized frames. A goal-conditioned gated recurrent unit (GRU) encodes an eight-frame history of 20-dimensional motor, perception, and depth features and predicts three-axis motor increments together with an auxiliary task label. The best saved model contains 773,221 parameters and achieved a held-out action mean-squared error of 0.002084 and task accuracy of 100%. Layer 2 couples lumen geometry with a continuous progress–distance authority schedule, allowing bounded visual steering to increase gradually without overriding the global navigation direction. Layer 3 combines U-Net mucus segmentation, YOLO-based anatomical recognition, Depth Anything V2 distance estimation, continuously active vacuum suction, and progress-aware mission continuity. Integrated phantom trials completed target-directed navigation, online junction centering, secretion clearance, continuation of the remaining inspection, and autonomous homing. Quantitative route, collision, clearance, and timing results are organized in a locked reporting protocol and remain to be inserted from the final trial sheet. The principal innovation is a unified architecture in which learned global intent, geometry-driven local correction, anatomical state, and therapeutic interaction operate at explicitly bounded levels of authority.

Index Terms—Medical robots and systems; imitation learning; autonomous navigation; bronchoscopy; visual servoing; semantic segmentation.

I. INTRODUCTION

Robotic bronchoscopy improves stability and peripheral reach, with feasibility, multicenter, and shape-sensing studies demonstrating its clinical potential [1]–[5]. Meta-analyses nevertheless report heterogeneous performance and motivate standardized system-level evaluation [6], [7]. Most deployed platforms remain operator driven, while autonomous traversal must handle repeated visual patterns, uncertain lumen geometry, nonrigid contact, and secretions.

Recent bronchoscopy systems use simulator-trained shared control, multimodal reinforcement learning, topological localization, or pure-vision agents [8]–[12]. Their progress exposes a central control dilemma: a global policy is needed to preserve anatomical intent, but local image feedback is needed to reduce wall contact. Unconstrained image centering can select the wrong lumen or accumulate large steering bias. A practical system must therefore regulate when

[TODO: Insert authors, affiliations, funding, and corresponding-author details.]

FIGURE 1 PLACEHOLDER — Complete three-layer architecture

Draw five horizontal blocks: expert data and GRU training; RGB/YOLO/U-Net/Depth Anything perception; Layer 1 global policy; Layer 2 progress-weighted visual correction; Layer 3 mucus intervention and mission continuity. Show camera, M_0 – M_2 , continuously active vacuum pump, motor feedback, anatomical goal, saved progress index, and final homing arrow. Use different colors for learned models, deterministic safety logic, and physical hardware.

Fig. 1. The proposed multimodal AutoNav3Layer framework.

and how strongly visual evidence may modify the global command. It should also manage mucus during inspection without discarding route progress.

We propose a progress-aware three-layer architecture (Fig. 1). Its contributions are:

- 1) a compact goal-conditioned GRU trained from synchronized expert motor, visual, depth, and anatomical data, with directly auditable training metrics;
- 2) a multimodal perception stack combining U-Net target segmentation, YOLO anatomical recognition, topology- and motion-constrained temporal confirmation, and Depth Anything V2 metric depth;
- 3) a progress–distance weighted junction controller that inserts a bounded M_1/M_2 correction into the global command while preserving the insertion axis and long-horizon route direction;
- 4) a continuously active suction strategy coupled with mucus tracking, depth-aware approach, saved navigation state, mission continuation, and autonomous homing; and
- 5) an evaluation framework spanning model accuracy, route completion, centerline tracking, wall contact, mucus clearance, continuity, and optional bounded reinforcement-learning refinement.

II. RELATED WORK

A. Robotic Bronchoscopy and Continuum Control

Bronchoscope mechanics inherit the nonlinearities of tendon-driven continuum robots. Constant-curvature, tendon-routing, Cosserat-rod, and medical continuum robot studies provide the modeling foundation [13]–[17]. Friction, hysteresis, mounting variation, and wall interaction limit model-

TABLE I
HARDWARE AND RUNTIME CONFIGURATION

Component	Role	Final reported value
M_0	insertion	range $[-900, 0]^\circ$, max $80^\circ/\text{s}$
M_1	horizontal bending	range $[-170, 170]^\circ$, max $30^\circ/\text{s}$
M_2	vertical bending	range $[-500, 500]^\circ$, max $80^\circ/\text{s}$
RGB camera	endoscopic perception	[RESULT: resolution, measured fps]
Vacuum pump	continuous secretion suction	[RESULT: model, pressure/flow]
GPU/CPU	model inference	[RESULT: hardware and total latency]
Phantom	airway evaluation	[RESULT: model/material/scale]

only control. Deep visual servoing has thus been explored for intraluminal navigation [18]; bronchoscopy-specific systems additionally use CT-derived environments, images, or robot pose [8], [9], [19]. Our approach separates global anatomical intent from bounded local correction.

B. Imitation and Reinforcement Learning

Learning from demonstration transfers expert skill without unsafe unrestricted exploration [20], [21]. Behavioral cloning is data efficient but can suffer from distribution shift, motivating aggregation and adversarial formulations [22], [23]. Goal tokens disambiguate actions at shared observations [24], while LSTM and GRU models capture temporal context [25], [26]. Recent robot-learning studies emphasize demonstration quality, temporal action representations, multimodality, and closed-loop evaluation [27]–[33]. For conservative post-training refinement, we formulate a bounded PPO residual using the clipped policy objective [34] and a reproducible implementation framework [35].

C. Endoscopic Perception

U-Net and later segmentation architectures established strong dense-prediction baselines [36]–[38]. Foundation models improve transferable visual representations [39], [40], while real-time one-stage detection supports anatomical recognition [41]. Monocular depth and endoscopic adaptation provide geometric cues under limited texture [42]–[46]. Because airway suction must be bounded and clinically supervised, we also follow bronchoscope suction studies and airway management guidance [47], [48].

III. SYSTEM AND DATA

A. Platform

The system comprises a flexible bronchoscope, RGB endoscopic camera, three position-controlled motors, encoders, a vacuum suction pump, and a GPU workstation. M_0 controls insertion; M_1 and M_2 bend the distal tip in orthogonal directions. The pump remains active throughout autonomous inspection, so Layer 3 controls alignment and exposure rather than pump switching. Position, speed, image freshness, depth validity, and operator-stop conditions are checked before every command.

B. Expert Demonstrations

Manual expert operation was synchronized with images, motor angles and velocities, perception features, depth values, timestamps, anatomical targets, and motor increments. The curated set contains 87 complete trajectories and 30,730

TABLE II
GOAL-CONDITIONED EXPERT DATASET

Target	Trajectories	Frames	Mean frames
TR	5	640	128
LMB	23	8,196	356
RMB	26	6,326	243
LUB	8	3,344	418
LLB	8	6,071	759
RML	8	3,237	405
RLL	9	2,916	324
Total	87	30,730	353

FIGURE 2 PLACEHOLDER — U-Net training and segmentation

Panel (a): representative RGB/mask pairs for clean lumen, thin mucus, thick mucus, specular highlight, blur, and partial occlusion. Panel (b): VGG-U-Net encoder/decoder diagram. Panel (c): train/validation loss and Dice curves. Panel (d): test examples with TP/FP/FN overlays. Include dataset sizes and 95% CI for Dice and IoU.

Fig. 2. Mucus segmentation training and held-out performance.

frames over seven recorded targets (Table II). Splitting is performed by whole trajectory to prevent adjacent-frame leakage.

IV. MULTIMODAL PERCEPTION

A. U-Net Mucus Segmentation

The deployed segmentation branch uses a VGG-encoder U-Net with skip-connected decoder blocks and a 1×1 output projection. RGB frames are resized to 512×512 . The current inference configuration separates mucus from background; a three-class training variant additionally represents the instrument. Training uses cross-entropy plus soft Dice loss,

$$\mathcal{L}_{seg} = \mathcal{L}_{CE} + \lambda_D(1 - \text{Dice}), \quad (1)$$

Adam with initial learning rate 10^{-4} , cosine decay, and batch size two. Geometric and photometric augmentation should be reported exactly from the training run. The largest valid connected mucus component supplies centroid $\mathbf{c}_t^m$, area A_t^m , and mask M_t^m .

B. YOLO Anatomical Recognition and Temporal Confirmation

A lightweight nine-class YOLO detector recognizes TR, RMB, RUL, RML, RLL, LMB, LUB, LLB, and BI. The model was initialized from YOLO11n and trained for 120 epochs at 640×640 , batch size 16, AdamW learning rate 5×10^{-4} , cosine decay, and weak geometric/color augmentation. It contains 2.59M parameters. Raw detections are filtered at confidence 0.30 and passed to a motion-aware

TABLE III
VERIFIED YOLO VALIDATION METRICS

Classes	Precision	Recall	mAP50	mAP50-95
9	0.8926	0.7797	0.9096	0.6006

FIGURE 3 PLACEHOLDER — Multimodal perception results

Use four columns for raw RGB, U-Net mucus mask, YOLO anatomical detection, and Depth Anything map. Add a topology panel showing raw YOLO label, candidate counter, confirmed label, insertion/withdrawal direction, and permitted tree transition. Add a depth-calibration scatter plot with fitted line and error.

Fig. 3. U-Net, YOLO, topology, and depth outputs in representative airway scenes.

anatomical state estimator. Only the current node, an allowed child during insertion, or the parent during withdrawal can be confirmed. Candidates require repeated observations; evidence is frozen during stationary frames, and branch-specific windows use insertion progress and bending evidence. This prevents single-frame anatomical jumps and makes location estimates consistent with the bronchial tree.

C. Depth Anything V2 and Robust Target Depth

Metric Depth Anything V2 estimates a dense depth map D_t . For a target mask, we erode boundary pixels, reject invalid samples, compute a robust central depth, apply scale/bias calibration, reject large temporal jumps, and filter with an EMA coefficient of 0.2. The resulting depth $\bar{z}_t$ controls insertion: approach when the target is distant, stop inside a safety band, and withdraw when too close. Depth is used as a control cue rather than an uncalibrated absolute measurement; phantom calibration error must be reported.

V. HIERARCHICAL CONTROL

A. Layer 1: Goal-Conditioned Recurrent Imitation

Each observation $\mathbf{o}_t \in \mathbb{R}^{20}$ contains mucus and junction features, center/mean/maximum depth, three directional path depths, three motor angles, three motor velocities, and elapsed time. An MLP maps it through $20 \rightarrow 128 \rightarrow 128$ with LayerNorm and ELU. The anatomical goal is embedded in 16 dimensions and concatenated across an eight-frame window:

$$\mathbf{e}_i = f_o(\mathbf{o}_i), \quad \mathbf{u}_i = [\mathbf{e}_i \| \text{Embed}(g)], \quad (2)$$

$$\mathbf{h}_i = \text{GRU}(\mathbf{u}_i, \mathbf{h}_{i-1}), \quad i = t-7, \dots, t. \quad (3)$$

A two-layer GRU has hidden size 256 and dropout 0.1. Its action head is $256 \rightarrow 128 \rightarrow 3$ with ELU and tanh; the task head is $256 \rightarrow 64 \rightarrow 2$. Normalized actions are scaled by $[5, 3, 3]^\circ$ for $M_0/M_1/M_2$. The total training objective is

$$\mathcal{L}_{BC} = \|\hat{\mathbf{a}}_t - \mathbf{a}_t^E\|_2^2 + \lambda_s \text{CE}(\hat{\mathbf{z}}_t, s_t^E). \quad (4)$$

FIGURE 4 PLACEHOLDER — GRU structure and training curves

Left: exact tensor dimensions for the 20-D observation encoder, 16-D goal embedding, 8-frame GRU, action head, and task head. Right: train/validation navigation loss and task accuracy for all available runs; mark the selected epoch 29 model and report mean $\pm$ SD over at least three seeds.

Fig. 4. Goal-conditioned recurrent imitation policy and optimization history.

TABLE IV
VERIFIED GRU TRAINING EVIDENCE

Model	Parameters	Selected epoch	Val. action MSE	Val. task acc.
Goal-conditioned GRU	773,221	29	0.002084	1.000

Training uses AdamW, OneCycleLR, mixed precision, gradient norm clipping at 1.0, batch size 64, and a trajectory-wise 90/10 split. The best model is selected by validation navigation loss.

Layer 1 integrates the action into a bounded nominal target. To retain an auditable global direction under distribution shift, the controller constrains the learned command inside an expert-consistent envelope. The exact deployment coefficient and envelope width must be logged in the final experiment. This separation allows the learned policy to express goal-conditioned motor timing while preserving safe long-horizon intent.

B. Layer 2: Progress-Weighted Junction Correction

Dark-lumen candidates are extracted after circular masking and morphological opening. Among candidates exceeding the minimum area, the geometric center nearest the image center is selected. For normalized error $\mathbf{e}_t = [e_t^x, e_t^y]^\top$, the correction authority is the product of a progress weight and distance weight. With route progress $s_t \in [0, 1]$,

$$w_p(s_t) = \beta_0 + (1 - \beta_0) \min\left(\frac{s_t}{s_f}, 1\right)^2, \quad (5)$$

where $\beta_0 = 0.25$ and $s_f = 0.80$. For center distance $d_t = \|\mathbf{e}_t\|_2$,

$$w_d(d_t) = \begin{cases} 1, & d_t \leq 0.30, \\ (0.75 - d_t)/(0.75 - 0.30), & 0.30 < d_t < 0.75, \\ 0, & d_t \geq 0.75. \end{cases} \quad (6)$$

After an axis dead-zone of 0.08, raw offsets use gains $K_1 = 3.0$ and $K_2 = 2.5$, EMA coefficient 0.25, and per-axis total limit 5° . The command is

$$\mathbf{q}_t^{\text{cmd}} = \mathbf{q}_t^{\text{nom}} + w_p(s_t)w_d(d_t)[0, \delta q_{1,t}^v, \delta q_{2,t}^v]^\top. \quad (7)$$

Thus M_0 preserves Layer 1 progress, whereas M_1/M_2 receive small, smooth, nonaccumulating centerline corrections. Independent visual alignment uses a 10-pixel dead-zone

FIGURE 5 PLACEHOLDER — Layer 2 correction mechanism

Panel (a): multiple lumen candidates and nearest-center selection. Panel (b): $w_p(s)$ and $w_d(d)$ curves. Panel (c): Layer 1 nominal M_1/M_2 , bounded visual offsets, and final commands across a bifurcation. Panel (d): image-plane center error with/without Layer 2 and synchronized wall-contact markers.

Fig. 5. Continuous progress–distance weighting preserves global intent while improving local centering.

FIGURE 6 PLACEHOLDER — Layer 3 sequence and state variables

Show six real frames: cruising with pump on, mucus detection, depth-aware approach, centered suction, 15-frame absence confirmation, and continued route. Below, plot route index, M_0 – M_2 , mask area, depth, target error, pump state, and final homing. Mark the saved pose and maximum return error.

Fig. 6. Continuous suction, mucus tracking, progress-aware mission continuity, and homing.

and a far-fast/near-slow speed curve; motor-axis signs and vertical scaling were calibrated on the physical robot. This structure is related to bounded residual control [49], but uniquely modulates authority with procedural progress and visual geometry.

C. Layer 3: Continuous Suction and Mission Continuity

The vacuum pump remains active during the inspection. When the U-Net mask exceeds the valid-area threshold, the controller stores route progress k^* and motor configuration $\mathbf{q}^*$, then prioritizes mucus tracking. M_1/M_2 center the mask, while M_0 approaches according to filtered depth and stops within the safety band. Missing detections do not immediately end the intervention: the target must remain absent for $N_{clear} = 15$ frames. This dropout-tolerant rule, together with mask-area evolution and depth validity, suppresses false completion caused by blur or specular highlights.

After clearance, the robot returns to $\mathbf{q}^*$ within a 5° maximum axis error, re-enters the remaining route at progress index k^* , completes the selected airway inspection, and finally drives all three navigation axes to the defined origin. The hierarchy is therefore

$$\begin{aligned} \text{NAVIGATE} &\rightarrow \text{TRACK/CLEAR} \rightarrow \text{SAVED POSE} \\ &\rightarrow \text{CONTINUE} \rightarrow \text{HOME.} \end{aligned} \quad (8)$$

Continuous suction simplifies actuator coordination but requires reporting pump pressure, total exposure time, and any mucosal-contact surrogate.

FIGURE 7 PLACEHOLDER — Phantom protocol

Photograph the robot, fixed start jig, airway phantom, target branches, vacuum pump and collection bottle. Overlay predefined mucus positions and perturbation conditions. Add a randomized trial timeline and failure taxonomy.

Fig. 7. Standardized hardware evaluation setup.

D. Bounded Reinforcement-Learning Refinement

We formulate PPO as an optional residual tuner rather than an unrestricted motor policy. Its state contains GRU hidden features, route progress, center error, depth, tracking error, and recent contact evidence. Its action adjusts only Layer 2 gain and offset within fixed bounds. A suitable reward is

$$\begin{aligned} r_t = & w_g \Delta s_t - w_c C_t - w_e \|\mathbf{e}_t\|_2 - w_j \|\Delta \mathbf{a}_t\|_2^2 \\ & - w_b \|\delta \mathbf{q}_t^v\|_2^2 + w_T \mathbb{I}_{goal}, \end{aligned} \quad (9)$$

where C_t is contact. Domain randomization covers start pose, illumination, friction, mask noise, and depth scale. The learned residual is accepted only when it improves held-out simulated success and then passes the same hardware limits. The existing standalone PPO artifact is not sufficient for this claim: its logged evaluation success is 0%, so the AutoNav-specific PPO row remains a required experiment rather than a reported improvement.

VI. EXPERIMENTS AND RESULTS

A. Study Design

Experiments should use paired, randomized trials on the same airway phantom. Targets include at least LMB, RMB, LUB, LLB, RUL, RML, and RLL. Conditions are nominal start, start-pose perturbation, illumination variation, lens contamination, and mucus placement at predefined locations. Report sample size per target, failure taxonomy, 95% confidence intervals, and paired tests with multiple-comparison correction.

Baselines are: (B0) expert manual control; (B1) frame-wise BC; (B2) GRU without goal; (B3) goal-conditioned GRU/Layer 1; (B4) reactive visual tracking only; (B5) Layer 1+2; (B6) full Layer 1+2+3; and (B7) full system with validated PPO residual, if available. Metrics are route success, wrong-branch rate, centerline error, contact count/duration, completion time, tracking error, intervention count, mucus clearance, clearance time, return-to-saved-pose error, continued route success, and homing error.

B. Model-Level Results

Table IV and Table III contain values recovered directly from saved model artifacts. The selected GRU model reduced validation action MSE from 0.02577 in the first logged epoch to 0.002084 at epoch 29, while task accuracy remained 100% on the recorded validation split. Because task labels may be

TABLE V
PERCEPTION MODEL RESULTS TO REPORT

Model	Task	Precision	Recall	Dice/mAP	Latency
VGG-U-Net	mucus mask	[RESULT: -]	[RESULT: -]	[RESULT: Dice/IoU]	[RESULT: ms]
YOLO11n	anatomy	0.8926	0.7797	0.9096 mAP50	[RESULT: ms]
Depth Anything V2	depth	-	-	[RESULT: MAE/RMSE]	[RESULT: ms]
Fusion stack	synchronized	-	-	[RESULT: valid-frame rate]	[RESULT: fps]

TABLE VI
NAVIGATION AND JUNCTION-CENTERING COMPARISON

Method	Success	Wrong branch	Center error	Contacts	Time
Expert	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]
Frame BC	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]
Layer 1	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]
Reactive vision	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]
Layer 1+2	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]

easy to separate, macro-F1 and per-class confusion must also be reported. The YOLO detector reached mAP50 0.9096, but recall 0.7797 indicates that temporal confirmation must tolerate missed frames. U-Net and depth results cannot be completed until the held-out masks and calibration logs are restored.

C. Integrated Results

Qualitative integrated tests completed the full sequence of target-directed navigation, progress-weighted junction correction, continuous mucus suction, mission continuation, route completion, and homing. The final paper must replace this statement with the quantitative trial matrix below; videos alone are not sufficient evidence.

D. Software Verification

The earlier control revision passed 92 hardware-independent logical checks. The current anatomy-recognition interface additionally passes 20 logic tests covering motor-direction gating, stale feedback, paused-state evidence, anatomical topology, latest-frame transport, detection/recording separation, and navigation-menu behavior. Layer 2 also has dedicated finite-offset and long-duration tests; one source-consistency assertion currently expects an older vertical speed scale (4.0 rather than the tuned 2.0) and must be updated before the paper claims a fully clean regression suite. These checks are not physical trials and are excluded from efficacy statistics.

VII. DISCUSSION

The system’s main novelty is not any single network, but authority-aware fusion. The GRU captures goal-conditioned temporal motor behavior; YOLO maintains anatomical context; U-Net and depth supply local therapeutic geometry; and Layer 2 inserts only a bounded correction whose strength depends on route progress and visual distance. This avoids two common failure modes: a local controller overriding the intended branch and a global controller ignoring current wall geometry.

The current model evidence also reveals limitations. GRU task accuracy of 100% may reflect an easy label distribution and needs macro-F1 and target-stratified testing. YOLO recall below 0.8 justifies temporal confirmation but may delay

TABLE VII
ABLATION OF LAYER 2 OPTIMIZATION

Variant	Center error	Contacts	Max offset	Success
Largest lumen	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]
Nearest lumen	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]
+ dead-zone/EMA	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]
+ progress weight	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]
Full bounded method	[RESULT: -]	[RESULT: -]	$\leq 5^\circ$	[RESULT: -]

TABLE VIII
MUCUS INTERVENTION AND MISSION CONTINUITY

Method	Clear success	Clear time	Pose error	Continue success	Home error
Manual suction	[RESULT: -]	[RESULT: -]	-	[RESULT: -]	[RESULT: -]
Tracking, no depth	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]
Tracking + depth	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]
Full Layer 3	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]	[RESULT: -]

anatomical transitions. U-Net metrics and depth calibration are missing from the repository. Continuous suction reduces switching complexity but requires pressure, exposure, fluid-removal, and tissue-safety reporting. Finally, the standalone PPO run did not solve its evaluation task; reinforcement learning must not be credited with system improvement until the bounded AutoNav residual passes both simulation and hardware ablations.

Future work will expand expert data to missing/rare goals, collect multi-operator demonstrations, train U-Net with hard negatives and temporal consistency, calibrate depth across airway materials, learn uncertainty-aware fusion, and evaluate on ex-vivo and in-vivo models. Clinical translation will additionally require sterility, suction-pressure safety, failure containment, and supervised human takeover protocols.

VIII. CONCLUSION

We introduced a progress-aware three-layer framework for autonomous robotic bronchoscopy. A goal-conditioned GRU learns temporal expert control; multimodal perception combines U-Net mucus masks, YOLO anatomical recognition, and Depth Anything geometry; a bounded progress-weighted visual controller improves junction centering; and continuous suction is integrated with mucus tracking, mission continuity, and homing. Saved artifacts provide quantitative GRU and YOLO evidence, while integrated trials establish qualitative feasibility of the complete procedure. The remaining tables explicitly define the quantitative evidence required for an IEEE RA-L submission.

REFERENCES

- [1] D. I. K. Fielding, F. Bashirzadeh, J. H. Son, M. Todman, A. Chin, L. Tan, K. Steinke, M. N. Windsor, and A. W. Sung, “First human use of a new robotic-assisted fiber optic sensing navigation system for small peripheral pulmonary nodules,” *Respiration*, vol. 98, no. 2, pp. 142–150, 2019.
- [2] U. Chaddha, S. P. Kovacs, C. Manley, D. K. Hogarth, G. Cumbo-Nacheli, S. V. Bhavani, R. Kumar, M. Shende, J. P. Egan, and S. Murgu, “Robot-assisted bronchoscopy for pulmonary lesion diagnosis: Results from the initial multicenter experience,” *BMC Pulmonary Medicine*, vol. 19, p. 243, 2019.
- [3] J. R. Rojas-Solano, L. Ugalde-Gamboa, and M. Machuzak, “Robotic bronchoscopy for diagnosis of suspected lung cancer: A feasibility study,” *Journal of Bronchology and Interventional Pulmonology*, vol. 25, no. 3, pp. 168–175, 2018.

TABLE IX
REINFORCEMENT-LEARNING REFINEMENT

Policy	Sim. success	Hardware success	Contact reduction
Standalone prototype	0%	not tested	not established
Bounded AutoNav PPO	[RESULT: -]	[RESULT: -]	[RESULT: -]

FIGURE 8 PLACEHOLDER — Final quantitative results

Panel (a): route success by target and method with 95% CI. Panel (b): paired center error and contact duration for Layer 1 vs Layer 1+2. Panel (c): mucus area/depth/centering through Layer 3. Panel (d): saved-pose error, continued route success and homing error. Panel (e): ablation waterfall.

Fig. 8. Required final system-level results and ablations.

- [4] A. C. Chen, N. J. Pastis, A. K. Mahajan, S. J. Khandhar, M. J. Simoff, M. S. Machuzak, J. Cienia, T. R. Gildea, and G. A. Silvestri, "Robotic bronchoscopy for peripheral pulmonary lesions: A multicenter pilot and feasibility study (BENEFIT)," *Chest*, vol. 159, no. 2, pp. 845–852, 2021.
- [5] O. Kalchiem-Dekel, J. G. Connolly, I.-H. Lin, B. C. Husta, P. S. Adusumilli, J. A. Beattie, D. J. Buonocore, J. D. Dyco, P. Fuentes, D. R. Jones, R. P. Lee, B. J. Park, G. Rocco, M. Chawla, and M. J. Bott, "Shape-sensing robotic-assisted bronchoscopy in the diagnosis of pulmonary parenchymal lesions," *Chest*, vol. 161, no. 2, pp. 572–582, 2022.
- [6] M. S. Ali, U. K. Ghorri, M. T. Wayne, E. Shostak, and J. D. Cardenas, "Diagnostic performance and safety profile of robotic-assisted bronchoscopy: A systematic review and meta-analysis," *Annals of the American Thoracic Society*, vol. 20, no. 12, pp. 1801–1812, 2023.
- [7] C. Zhang, F. Xie, R. Li, N. Cui, F. J. F. Herth, and J. Sun, "Robotic-assisted bronchoscopy for the diagnosis of peripheral pulmonary lesions: A systematic review and meta-analysis," *Thoracic Cancer*, vol. 15, no. 7, pp. 505–512, 2024.
- [8] J. Zhang, L. Liu, P. Xiang, Q. Fang, X. Nie, H. Ma, J. Hu, R. Xiong, Y. Wang, and H. Lu, "AI co-pilot bronchoscope robot," *Nature Communications*, vol. 15, p. 241, 2024.
- [9] J. Zhao, H. Chen, Q. Tian, J. Chen, B. Yang, and H. Liu, "BronchoCopilot: Towards autonomous robotic bronchoscopy via multi-modal reinforcement learning," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems*, 2024.
- [10] K. M. Cold, Z. Cheng, L. Konge, A. O. Nielsen, T. R. Savarimuthu, and B. M. G. Oliveira, "An autonomous bronchoscopy robot controlled by artificial intelligence (BronchoBot) outperforms experienced bronchoscopists in a simulated setting," *ERJ Open Research*, vol. 12, no. 2, pp. 00970–2025, 2026.
- [11] J. Wu, M. Luo, F. Xie, M. Zhang, H. Zhang, C. Zhang, J. Wang, J. Sun, Y. Gu, and G.-Z. Yang, "Long-short term agents for pure-vision bronchoscopy robotic autonomy," 2026.
- [12] C. Tomasini, L. Riazuelo, and A. C. Murillo, "Online topological localization for navigation assistance in bronchoscopy," 2025.
- [13] D. B. Camarillo, C. F. Milne, C. R. Carlson, M. R. Zinn, and J. K. Salisbury, "Mechanics modeling of tendon-driven continuum manipulators," *IEEE Transactions on Robotics*, vol. 24, no. 6, pp. 1262–1273, 2008.
- [14] R. J. W. III and B. A. Jones, "Design and kinematic modeling of constant curvature continuum robots: A review," *The International Journal of Robotics Research*, vol. 29, no. 13, pp. 1661–1683, 2010.
- [15] D. C. Rucker, R. J. W. III, G. S. Chirikjian, and N. J. Cowan, "Statics and dynamics of continuum robots with general tendon routing and external loading," *IEEE Transactions on Robotics*, vol. 27, no. 6, pp. 1033–1044, 2011.
- [16] J. Burgner-Kahrs, D. C. Rucker, and H. Choset, "Continuum robots for medical applications: A survey," *IEEE Transactions on Robotics*, vol. 31, no. 6, pp. 1261–1280, 2015.
- [17] J. Till, V. Aloï, and D. C. Rucker, "Real-time dynamics of soft and continuum robots based on cosserat rod models," *The International Journal of Robotics Research*, vol. 38, no. 6, pp. 723–746, 2019.
- [18] J. F. Lazo, C.-F. Lai, S. Moccia, B. Rosa, M. Catellani, M. de Mathelin, G. Ferrigno, P. Breedveld, J. Dankelman, and E. D. Momi, "Autonomous intraluminal navigation of a soft robot using deep-learning-based visual servoing," 2022.
- [19] J. Xu, B. Guan, J. Zhao, B. Yi, and J. Li, "LungDepth: Self-supervised multi-frame monocular depth estimation for bronchoscopy," *The International Journal of Medical Robotics and Computer Assisted Surgery*, vol. 21, no. 1, p. e70050, 2025.
- [20] B. D. Argall, S. Chernova, M. Veloso, and B. Browning, "A survey of robot learning from demonstration," *Robotics and Autonomous Systems*, vol. 57, no. 5, pp. 469–483, 2009.
- [21] T. Osa, J. Pajarinen, G. Neumann, J. A. Bagnell, P. Abbeel, and J. Peters, "An algorithmic perspective on imitation learning," *Foundations and Trends in Robotics*, vol. 7, no. 1–2, pp. 1–179, 2018.
- [22] S. Ross, G. J. Gordon, and J. A. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proc. Int. Conf. Artificial Intelligence and Statistics*, ser. PMLR, vol. 15, 2011, pp. 627–635.
- [23] J. Ho and S. Ermon, "Generative adversarial imitation learning," in *Advances in Neural Information Processing Systems*, vol. 29, 2016.
- [24] F. Codevilla, M. Muller, A. Lopez, V. Koltun, and A. Dosovitskiy, "End-to-end driving via conditional imitation learning," in *Proc. IEEE Int. Conf. Robotics and Automation*, 2018, pp. 4693–4700.
- [25] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [26] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in *Proc. Conf. Empirical Methods in Natural Language Processing*, 2014, pp. 1724–1734.
- [27] A. Mandlekar, D. Xu, J. Wong, S. Nasiriany, C. Wang, R. Kulkarni, L. Fei-Fei, S. Savarese, Y. Zhu, and R. Martin-Martin, "What matters in learning from offline human demonstrations for robot manipulation," in *Proc. Conf. Robot Learning*, ser. PMLR, vol. 164, 2022, pp. 1678–1690.
- [28] P. Florence, C. Lynch, A. Zeng, O. A. Ramirez, A. Wahid, L. Downs, A. Wong, J. Lee, I. Mordatch, and J. Tompson, "Implicit behavioral cloning," in *Proc. Conf. Robot Learning*, ser. PMLR, vol. 164, 2022, pp. 158–172.
- [29] N. M. M. Shafiullah, Z. Cui, A. Altanzaya, and L. Pinto, "Behavior transformers: Cloning k modes with one stone," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 22955–22968.
- [30] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *The International Journal of Robotics Research*, 2024.
- [31] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," in *Robotics: Science and Systems*, 2023.
- [32] C. Lynch, M. Khansari, T. Xiao, V. Kumar, J. Tompson, S. Levine, and P. Sermanet, "Learning latent plans from play," in *Proc. Conf. Robot Learning*, ser. PMLR, vol. 100, 2020, pp. 1113–1132.
- [33] S. Levine, C. Finn, T. Darrell, and P. Abbeel, "End-to-end training of deep visuomotor policies," *Journal of Machine Learning Research*, vol. 17, no. 39, pp. 1–40, 2016.
- [34] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017.
- [35] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-baselines3: Reliable reinforcement learning implementations," *Journal of Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.
- [36] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention*, 2015, pp. 234–241.
- [37] K. He, G. Gkioxari, P. Dollar, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Computer Vision*, 2017, pp. 2961–2969.
- [38] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *European Conf. Computer Vision*, 2018, pp. 801–818.

- [39] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. Berg, W.-Y. Lo, P. Dollar, and R. Girshick, "Segment anything," in *Proc. IEEE/CVF Int. Conf. Computer Vision*, 2023, pp. 4015–4026.
- [40] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.
- [41] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition*, 2016, pp. 779–788.
- [42] C. Godard, O. M. Aodha, M. Firman, and G. Brostow, "Digging into self-supervised monocular depth estimation," in *Proc. IEEE/CVF Int. Conf. Computer Vision*, 2019, pp. 3828–3838.
- [43] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth anything V2," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.
- [44] K. B. Ozyoruk, G. I. Gokceler, T. L. Bobrow, G. Coskun, K. Incetan, Y. Almalioglu, F. Mahmood, E. Curto, L. Perdigoto, M. Oliveira, H. Sahin, H. Araujo, H. Alexandrino, N. J. Durr, H. B. Gilbert, and M. Turan, "Endoslam dataset and an unsupervised monocular visual odometry and depth estimation approach for endoscopic videos," *Medical Image Analysis*, vol. 71, p. 102058, 2021.
- [45] B. Cui, M. Islam, L. Bai, A. Wang, and H. Ren, "EndoDAC: Efficient adapting foundation model for self-supervised depth estimation from any endoscopic camera," 2024.
- [46] M. S. Zeinoddin, C. Lena, J. Qu, L. Carlini, M. Magro, S. Kim, E. D. Momi, S. Bano, M. Grech-Sollars, E. Mazomenos, D. C. Alexander, D. Stoyanov, M. J. Clarkson, and M. Islam, "DARES: Depth anything in robotic endoscopic surgery with self-supervised vector-LoRA of the foundation model," 2024.
- [47] C. R. Lamb, E. Yavarovich, V. Kang, E. L. Servais, L. B. Sheehan, S. Shadchahr, J. Weldon, M. J. Rousseau, and G. P. Tirrell, "Performance of a new single-use bronchoscope versus a marketed single-use comparator: A bench study," *BMC Pulmonary Medicine*, vol. 22, p. 189, 2022.
- [48] T. C. Blakeman, J. B. Scott, M. A. Yoder, E. Capellari, and S. L. Strickland, "AARC clinical practice guidelines: Artificial airway suctioning," *Respiratory Care*, vol. 67, no. 2, pp. 258–271, 2022.
- [49] T. Johannink, S. Bahl, A. Nair, J. Luo, A. Kumar, M. Loskyll, J. A. Ojea, E. Solowjow, and S. Levine, "Residual reinforcement learning for robot control," in *Proc. IEEE Int. Conf. Robotics and Automation*, 2019, pp. 6023–6029.